

CLASS 8 MATH

Advanced Mastery Workbook

Equations, Identities, Mensuration & 100 Olympiad Puzzles

Chapter 1: Rational Numbers, Squares & Cubes

SECTION A: CORE CONCEPTS

1. Find the smallest number by which 252 must be multiplied to get a perfect square.

2. Find the cube root of 13,824.

3. Find a Pythagorean triplet whose one member is 16.

4. Solve: $(\frac{5}{8})^{-7} \times (\frac{8}{5})^{-4}$.

5. Express 0.0000000000085 in standard scientific notation.

SECTION B: REAL WORLD APPLICATIONS

1. The area of a square field is 60025 square meters. Find the perimeter of the field.

2. A cubical box has a volume of 512 cubic centimeters. What is the length of its side?

3. A plant cell has a size of 0.00001275 m. Express this size in standard form.

Chapter 2: Linear Equations in One Variable

SECTION A: CORE CONCEPTS

1. Solve for x : $8x + 4 = 3(x - 1) + 7$.

2. Solve for m : $m - \frac{(m - 1)}{2} = 1 - \frac{(m - 2)}{3}$.

3. Solve the linear equation: $15(y - 4) - 2(y - 9) + 5(y + 6) = 0$.

4. If $\frac{x}{2} - \frac{1}{5} = \frac{x}{3} + \frac{1}{4}$, find x .

5. Two numbers are in the ratio 5:3. If they differ by 18, what are the numbers?

SECTION B: REAL WORLD APPLICATIONS

1. The sum of three consecutive multiples of 8 is 888. Find the multiples.

2. A grandfather is ten times older than his granddaughter. He is also 54 years older than her. Find their present ages.

3. The perimeter of a rectangular swimming pool is 154 meters. Its length is 2 meters more than twice its width. What are its dimensions?

Chapter 3: Algebraic Expressions & Identities

SECTION A: CORE CONCEPTS

1. Multiply the binomials: $(2x + 5)$ and $(4x - 3)$.

2. Simplify using identity: $(3x + 4y)^2 - (3x - 4y)^2$.

3. Factorize the expression: $14pq + 35pqr$.

4. Factorize using identity: $x^2 - 16x + 64$.

5. Divide: $(24x^2y^3 - 16x^3y^2)$ by $8x^2y^2$.

SECTION B: REAL WORLD APPLICATIONS

1. The length and breadth of a rectangle are $(3x + 4)$ and $(5x - 2)$. Write an expression for its area.

2. If the area of a square is $4x^2 + 12xy + 9y^2$, what is the expression for its side length?

3. A printer costs $(5a^2 + 2b)$ \$. If a school buys $(3a - b)$ printers, what is the total algebraic cost?

Chapter 4: Mensuration & Exponents

SECTION A: CORE CONCEPTS

1. Find the area of a trapezium whose parallel sides are 12 cm and 20 cm, and the distance between them is 8 cm.

2. The diagonals of a rhombus are 7.5 cm and 12 cm. Find its area.

3. Find the total surface area of a cylinder with radius 7 cm and height 10 cm.

4. What is the volume of a cuboid of dimensions 12 cm x 8 cm x 5 cm?

5. Find the curved surface area of a right circular cylinder of height 14 cm and base radius 5 cm.

SECTION B: REAL WORLD APPLICATIONS

1. A suitcase measures 80 cm x 48 cm x 24 cm. How many square meters of tarpaulin cloth is required to cover 100 such suitcases?

2. A cylindrical milk tank has a radius of 1.5 m and a length of 7 m. Find the maximum quantity of milk (in liters) it can store.

3. A road roller takes 750 complete revolutions to move once over to level a road. The diameter of the road roller is 84 cm and length is 1 m. Find the area of the road.

Multi-Concept Combined Practice

LEVEL 1: CORE PROFICIENCY (AVERAGE STUDENT) - SET 1

1. Find the square root of 7,056.

2. Solve the equation: $5t - 3 = 3t - 5$.

3. Multiply: $(a + 7)$ by $(b - 5)$.

4. Find the area of a rhombus whose diagonals are 10 cm and 8.2 cm.

5. Express 3.02×10^{-6} in usual form.

6. Find the cube of 15.

7. Factorize: $x^2 + 8x + 16$.

8. Convert the ratio 2:3 into a percentage.

9. Find the volume of a cube with side 9 cm.

10. Evaluate $(3^0 + 4^{-1}) \times 2^2$.

Multi-Concept Combined Practice

LEVEL 1: CORE PROFICIENCY (AVERAGE STUDENT) - SET 2

1. Find the cube root of 2,744.

2. Solve for x: $\frac{x}{3} + \frac{5}{2} = -\frac{3}{2}$.

3. Simplify: $(x + 3)(x - 3)$.

4. Find the height of a cylinder whose radius is 7 cm and total surface area is 968 sq cm.

5. What is the value of 5^{-3} ?

6. Calculate the simple interest on \$4000 at 5% for 2 years.

7. Factorize: $25m^2 - 16$.

8. A shirt with a marked price of \$40 is sold at a 15% discount. What is the sale price?

9. The area of a rectangle is 240 sq cm. If its length is 20 cm, find its perimeter.

10. Express 45,000,000 in standard form.

Multi-Concept Combined Practice

LEVEL 2: ADVANCED APPLICATION (TOP 10%) - SET 1

1. Find the smallest square number that is divisible by each of the numbers 4, 9, and 10.

2. Ages of two friends are in the ratio 5:7. Four years from now, the ratio of their ages will be 3:4. Find their present ages.

3. Using the identity $(x+a)(x+b) = x^2 + (a+b)x + ab$, find the product of 103 x 104.

4. A cuboidal box has dimensions 60 cm x 40 cm x 30 cm. How many small cubes with side 5 cm can be placed in the box?

5. Find the amount and compound interest on \$10,800 for 3 years at 12.5% per annum compounded annually.

6. Factorize: $p^2 - 10p + 21$.

7. If 15 workers can build a wall in 48 hours, how many workers will be required to do the same work in 30 hours?

8. Simplify: $(25 \times t^{-4}) / (5^{-3} \times 10 \times t^{-8})$.

9. A train is moving at a uniform speed of 75 km/h. How far will it travel in 20 minutes?

10. Find the area of a regular hexagon with a side length of 6 cm.

Multi-Concept Combined Practice

LEVEL 2: ADVANCED APPLICATION (TOP 10%) - SET 2

1. Evaluate the square root of 12.25.

2. Solve: $(3y + 4)/(2 - 6y) = -2/5$.

3. Expand using an identity: $(2x + 3y)^2$.

4. A 14m deep well with an inner diameter of 3m is dug, and the earth taken out is evenly spread to form an embankment of width 4m. Find the height of the embankment.

5. A washing machine was bought at \$3300 including a tax of 10%. Find the price before tax was added.

6. Divide $44(x^4 - 5x^3 - 24x^2)$ by $11x(x - 8)$.

7. A machine in a soft drink factory fills 840 bottles in 6 hours. How many bottles will it fill in 5 hours?

8. Find the value of m for which $5^m / 5^{-3} = 5^5$.

9. If the area of a trapezium is 34 cm^2 and the length of one of the parallel sides is 10 cm and its height is 4 cm. Find the length of the other parallel side.

10. Factorize the expression: $4y^2 - 12y + 9$.

Multi-Concept Combined Practice

LEVEL 3: LOGICAL REASONING (TOP 1%) - SET 1

1. Is 68600 a perfect square? If not, find the smallest multiple that is a perfect square and find its square root.

2. The digits of a two-digit number differ by 3. If the digits are interchanged and the resulting number is added to the original number, we get 143. What can be the original number?

3. Simplify: $(ab - bc)^2 + 2ab^2c$.

4. Water is pouring into a cuboidal reservoir at the rate of 60 liters per minute. If the volume of the reservoir is 108 cubic meters, find the number of hours it will take to fill the reservoir.

5. Find the compound interest on \$8000 for 1.5 years at 10% per annum, interest being compounded half-yearly.

6. Factorize completely: $a^4 - b^4$.

7. A garrison of 120 men has provisions for 30 days. After 5 days, 5 more men join them. How long will the remaining food last?

8. Evaluate $(8^{-1} \times 5^3) / 2^{-4}$.

9. A polyhedron has 20 faces and 12 vertices. How many edges does it have? (Use Euler's Formula).

10. If $x + \frac{1}{x} = 5$, find the value of $x^2 + \frac{1}{x^2}$.

Multi-Concept Combined Practice

LEVEL 3: LOGICAL REASONING (TOP 1%) - SET 2

1. Find the smallest number by which 8192 must be divided so that the quotient is a perfect cube.

2. The numerator of a rational number is less than its denominator by 3. If the numerator is increased by 2 and the denominator is increased by 5, the new number becomes $\frac{1}{2}$. Find the original rational number.

3. Show that $(4pq + 3q)^2 - (4pq - 3q)^2 = 48pq^2$.

4. The lateral surface area of a hollow cylinder is 4224 sq cm. It is cut along its height and formed a rectangular sheet of width 33 cm. Find the perimeter of rectangular sheet.

5. A scooter was bought at \$42,000. Its value depreciated at the rate of 8% per annum. Find its value after one year.

6. Factorize: $x^2 - y^2 + 2x + 1$.

7. If a box of sweets is divided among 24 children, they will get 5 sweets each. How many would each get, if the number of the children is reduced by 4?

8. Simplify and write in exponential form: $(3^{-7} / 3^{-10}) \times 3^{-5}$.

9. What is the probability of picking a prime number from a standard 6-sided die?

10. Solve for x: $(2x - 1)/3 - (6x - 2)/5 = 1/3$.

Multi-Concept Combined Practice

LEVEL 4: CRITICAL THINKING (TOP 0.1%) - SET 1

1. Evaluate $\sqrt{0.00059049}$.

2. A man leaves half his property to his wife, one-third to his son, and the remaining \$10000 to his daughter. What is his total property?

3. If $x - \frac{1}{x} = 4$, find the value of $x^4 + \frac{1}{x^4}$.

4. A metallic cylinder of radius 8 cm and height 2 cm is melted and converted into a right circular cone of height 6 cm. Find the radius of the base of this cone.

5. Calculate the difference between the compound interest and the simple interest on \$20,000 for 2 years at 8% per annum.

6. Factorize the polynomial: $p^4 - 81q^4$.

7. Two pipes can fill a tank in 10 hours and 12 hours respectively while a third pipe empties the full tank in 20 hours. If all three pipes operate simultaneously, in how much time will the tank be filled?

8. Solve for n : $2^n + 2^{(n-1)} = 24$.

9. Find the number of sides of a regular polygon whose each interior angle is 165° .

10. If 15% of $x = 20\%$ of y , find $x:y$.

Multi-Concept Combined Practice

LEVEL 4: CRITICAL THINKING (TOP 0.1%) - SET 2

1. Find the cube root of 0.000001331.

2. Two years ago, a father was five times as old as his son. Two years later, his age will be 8 more than three times the age of the son. Find the present ages of father and son.

3. If $a + b + c = 9$ and $ab + bc + ca = 26$, find the value of $a^2 + b^2 + c^2$.

4. The internal measures of a cuboidal room are 12m x 8m x 4m. Find the total cost of whitewashing all four walls of a room, if the cost of whitewashing is \$5 per square meter.

5. A population of a town grows at 5% per annum. If its present population is 44,100, what was its population 2 years ago?

6. Divide $x^4 + x^2 + 1$ by $x^2 - x + 1$.

7. A can do a piece of work in 15 days and B alone can do it in 10 days. B works at it for 5 days and then leaves. A alone can finish the remaining work in how many days?

8. Find the value of x if $8^{(x+1)} = 64$.

9. The pie chart shows the breakdown of a \$50,000 budget. If the sector for 'Marketing' has a central angle of 72° , how much is spent on Marketing?

10. Solve the equation: $0.16(5x - 2) = 0.4x + 7$.

Multi-Concept Combined Practice

LEVEL 5: NATIONAL OLYMPIAD (TOP 10) - SET 1

1. If the square of a number ends in 9, what are the possible last digits of its cube?

2. In a two-digit number, the ten's digit is three times the unit's digit. When the number is decreased by 54, the digits are reversed. Find the number.

3. If $x^2 + 1/x^2 = 27$, find the value of $x - 1/x$.

4. A solid metallic sphere of radius 10.5 cm is melted and recast into a number of smaller cones, each of radius 3.5 cm and height 3 cm. Find the number of cones so formed.

5. A sum of money doubles itself at compound interest in 15 years. In how many years will it become eight times?

6. Factorize: $x^4 + 4$.

7. A, B, and C can do a work in 12, 15, and 20 days respectively. They started the work together. A left after 2 days and B left after 3 days of starting the work. In how many days will the remaining work be completed by C?

8. Evaluate: $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$ to infinity.

9. What is the maximum number of acute angles a convex polygon with 10 sides can have?

10. Find the unit digit of 7^{2026} .

Multi-Concept Combined Practice

LEVEL 5: NATIONAL OLYMPIAD (TOP 10) - SET 2

1. Find the smallest positive integer n such that $2026 \times n$ is a perfect cube.

2. A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km downstream. Determine the speed of the stream and that of the boat in still water.

3. If $a + b = 5$ and $ab = 6$, find the value of $a^3 + b^3$.

4. Three solid metallic cubes of sides 3 cm, 4 cm, and 5 cm are melted to form a single solid cube. Find the edge of the new cube.

5. The difference between compound interest and simple interest on a certain sum for 3 years at 10% per annum is \$93. Find the sum.

6. Factorize the algebraic expression: $x^2 - y^2 - z^2 + 2yz$.

7. Two trains are running in opposite directions with the same speed. If the length of each train is 120 meters and they cross each other in 12 seconds, find the speed of each train in km/hr.

8. Simplify: $\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \dots + \frac{1}{\sqrt{99}+\sqrt{100}}$.

9. In a regular polygon, the ratio of the interior angle to the exterior angle is 5:1. Find the number of sides.

10. If $3^x - 3^{(x-1)} = 18$, find the value of x^x .
